

SCIENTIFIC INVESTIGATIONS

Contemporary diagnostic visual and automated polysomnographic REM sleep without atonia thresholds in isolated REM sleep behavior disorder

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Study Objectives: Accurate diagnosis of isolated rapid eye movement (REM) sleep behavior disorder (iRBD) is crucial due to its injury potential and neurological prognosis. We aimed to analyze visual and automated REM sleep without atonia (RSWA) diagnostic thresholds applicable in varying clinical presentations in a contemporary cohort of patients with iRBD using submental (SM) and individual bilateral flexor digitorum superficialis (FDS) and anterior tibialis electromyography limb recordings during polysomnography.

Methods: We analyzed RSWA in 20 patients with iRBD and 20 age-, REM-, apnea-hypopnea index-matched controls between 2017 and 2022 for phasic burst durations, density of phasic, tonic, and “any” muscle activity (number of 3-second mini-epochs containing phasic or tonic muscle activity divided by the total number of REM sleep 3-second mini-epochs), and automated Ferri REM atonia index (RAI). Group RSWA metrics were comparatively analyzed. Receiver operating characteristic curves determined optimized area under the curve (AUC) and maximized specificity and sensitivity diagnostic iRBD RSWA thresholds.

Results: All mean RSWA metrics were higher in patients with iRBD than in controls ($P < .05$), except for selected anterior tibialis measures. Optimized, maximal specificity AUC diagnostic cutoffs for coprimary outcomes were: SM “any” 6.5%, 14.0% (AUC = 92.5%) and combined SM+FDS “any” 15.1%, 27.4% (AUC = 95.8%), while SM burst durations were 0.72, and 0.72 seconds (AUC 90.2%) and FDS RAI = 0.930, 0.888 (AUC 92.8%).

Conclusions: This study provides evidence for current quantitative RSWA diagnostic thresholds in chin and individual 4 limb muscles applicable in different iRBD clinical settings and confirms the key value of SM or SM+FDS to assure accurate iRBD diagnosis. Evolving iRBD recognition underscores the necessity of continuous assessment with future large, prospective, well-harmonized, multicenter polysomnographic analyses.

Keywords: REM sleep without atonia, isolated REM sleep behavior disorder, phasic muscle burst, automated analysis, computational measures, diagnosis, threshold cutoffs

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BRIEF SUMMARY

Current Knowledge/Study Rationale: REM sleep without atonia is required for accurate diagnosis of REM sleep behavior disorder, a prodromal synucleinopathy. Manually and automatically scored muscle activity measures in chin and 4 limbs have been proposed as a diagnostic threshold nearly a decade ago.

Study Impact: Using manual and automated methods, we provided current quantitative REM sleep without atonia thresholds (muscle activation density, burst duration, REM atonia index) in chin and individual 4 limb muscles, applicable in various clinical contexts. This study also confirms the importance of submental and/or flexor digitorum superficialis muscles in REM sleep behavior disorder diagnosis accuracy.

INTRODUCTION

Rapid eye movement (REM) sleep behavior disorder (RBD) is a parasomnia characterized by dream-enactment behaviors and excessive muscle activity in REM sleep (REM sleep without atonia [RSWA]), its polysomnographic (PSG) signature.¹ Early detection of isolated RBD is crucial, given its potentially injurious consequences and its significance as a prodromal parkinsonism state presaging evolution of overt neurodegenerative disease when presenting in older adults.^{2–4} Recently, RSWA has also emerged

as a biomarker for the alpha-synucleinopathies, having been shown to (1) aid distinction of presumed alpha-synucleinopathy cognitive impairment and dementia neurodegenerative disease phenotypes even when clinical dream-enactment behaviors are absent,⁵ (2) progress over time and associate with longer RBD symptom and disease duration,^{6,7} and (2) associate with more malignant motor, cognitive, and autonomic impairments in prodromal and manifest synucleinopathies.^{7–11}

Since 1992 when Lapiere and Montplaisir developed the original visual scoring approach, based on the density of the

presence of electromyography (EMG) activity per unit of time, numerous visual and automatic quantification methods for determining RSWA have been proposed, with various methods demonstrating comparable diagnostic accuracy in a recent meta-analysis.^{12–28} The International RBD Study Group Neurophysiology Working Group recently reviewed methods of RSWA quantification and made recommendations for the accurate polysomnographic diagnosis of RBD.²⁹ Three types of muscle activity have been defined and most frequently analyzed across different groups and methods: phasic, tonic and “any” (meaning phasic and/or tonic muscle activity, hereafter used without quotes for readability), with some differences in scoring criteria used.^{13,15,18} These measures have been based on the presence or absence of EMG activation over a defined time period (typically 3-second mini-epochs). In addition, our group has also shown the utility of phasic muscle activity burst durations as an additional valuable RSWA measure in RBD diagnosis, distinguishing isolated RSWA abnormalities from normative REM sleep muscle activity durations,³⁰ and characterizing synucleinopathy populations.^{5,15,31–33}

Insofar as recording site is concerned, quantification of any EMG activity in chin and phasic EMG activity in the bilateral flexor digitorum superficialis (FDS) has shown the best predictive value either with manual or automatic methods but, to date, has been previously analyzed for RBD diagnostic accuracy only by the Sleep Innsbruck Barcelona (SINBAR) group.^{34–36} Alternative muscles analyzed include any, tonic, and phasic chin EMG activity, in 3-second or 30-second epochs, or chin and FDS EMG activity in 30-second epochs using the method described by the American Academy of Sleep Medicine (AASM).²⁹

In the previous decade, quantification studies proposed several diagnostic RSWA thresholds for isolated RBD (iRBD), either based on different approaches to case-control comparisons, or defining levels exceeding 95th percentile thresholds in sleep center or community populations without RBD.^{15,30–32,34,37,38} Thus, chin tonic activity threshold has been found to range from 0.7% to 30% in 20-second or 30-second epochs scoring, chin any activity in 3-second epochs from 18.2% to 21.6%, and SINBAR metric (ie, combined chin any and FDS phasic) in 3-second epochs from 10.7% to 31.9%.^{15,31,34,37}

Accurate RSWA quantification appears to be the most sensitive parameter in RBD diagnosis in comparison to RBD history or video-PSG–documented behaviors.³⁹ On the other hand, the concept of prodromal RBD, comprising isolated RSWA with or without subtle covert motor and vocal behavioral events during sleep insufficient for RBD diagnosis, has emerged more recently, signaling a need for specific diagnostic RSWA criteria to avoid misdiagnosis.^{7,29,30,40,41}

Enhanced recognition of iRBD and RSWA suggests that accepted thresholds could evolve. Normative cohort analyses have shown that isolated RSWA is frequent among control populations without RBD, which could impact normative values and RBD thresholds in case-control design studies, as isolated RSWA frequency in different settings and populations has been estimated to range from 5.5% up to 25% or greater (depending on age and sex, and muscles analyzed).^{30,42} Moreover, temporal trends toward a more widespread recognition in the general

population could lead to earlier detection of iRBD and presentation with a milder phenotype and lower RSWA amounts, considering that higher RSWA levels parallel longer disease duration and greater pheno-conversion risk.^{6,32}

Our primary aim was to analyze visual and automated RSWA in a current cohort of patients with iRBD using submentalis (SM) and individual 4 limb EMG recordings in comparison to controls, to determine contemporary RSWA density thresholds applicable in diagnosis for different clinical settings. Secondary objectives were to determine optimal muscles and muscle combinations explaining variance in RSWA diagnostic thresholds; to confirm the application of a robust automatic RSWA quantification method for limb muscles, the Ferri REM Atonia Index (RAI)^{14,15,31,43}; and to explore the possibility of evolving temporal trends in RSWA diagnostic thresholds since our previous study.³¹

METHODS

Patients

A total of 40 patients (20 controls, 20 with iRBD) seen between 2017 and 2022 at the Mayo Clinic Center for Sleep Medicine were selected for retrospective RSWA analysis. All patients were free from antidepressant and any RSWA-modifying treatment. Patients with REM apnea-hypopnea index (AHI) > 15 events/h or a total REM time of < 5 minutes were excluded. A chart review was performed for age, sex, and other clinical and demographic factors. The patient group included 20 patients with iRBD meeting *International Classification of Sleep Disorders*, third edition (ICSD-3), RBD diagnosis criteria without any clinical signs of parkinsonism or dementia at the time of PSG.¹ The control group included 20 patients matched for age and REM AHI who were referred for evaluation of sleep-disordered breathing (SDB; n = 19) or positive airway pressure titration (n = 1), of whom 5 (25%) also had chronic insomnia and 4 (20%) had restless legs syndrome. All patients with iRBD were referred for PSG because of dream-enactment behavior, and in 13 (65%) SDB was also suspected. Eight (40%) patients with iRBD and 13 (65%, $P = .11$) controls underwent split-night sleep studies (which required an AHI ≥ 5 events/h during the first 2 hours of sleep, followed by a positive airway pressure titration trial), from which the second half of the night with low REM AHI was selected for analysis.¹⁵ The Mayo Clinic Institutional Review Board approved this study.

PSG recordings and sleep architecture analyses

Video PSG studies were recorded on a 31-channel Nicolet NicVue digital system (2017–2018, n = 24 studies) or a 33-channel Xtek Natus (Middleton, Wisconsin, USA) digital system thereafter with acquisition methods as previously described.¹⁵ The EMG recordings included SM and individual limb EMG recordings from the left anterior tibialis (AT), right AT, left FDS, and right FDS in both patients with RBD and controls. Sleep staging was performed on 30-second PSG epochs following AASM rules²⁷ and well-accepted standards for REM sleep state determination in the setting of RSWA.^{12,29} EMG channels

were displayed at a sensitivity of 10 μ V/mm with low- and high-frequency filters of 10 and 100 Hz, respectively, with a sampling rate of 500 Hz. Principal sleep architecture metrics were recorded for all patients from the total night (both diagnostic and treatment phases in case of split-night). SDB metrics (ie, AHI values for total sleep and REM sleep, following AASM 1A rule²⁷) were reported from the total night and condition scored phase to accurately indicate the severity of comorbid obstructive sleep apnea concomitant to RSWA scoring.

Analysis of RSWA

Visual quantitative analysis of EMG activity was performed using HypnoLab sleep scoring software (SWS, Inc, Colognola ai Colli, Italy). The overall, combined and single muscle (SM, FDS, and AT) tonic, phasic, and any (either tonic, phasic, or both forms of muscle activity occurring within the same mini-epoch) percent muscle activities were manually scored by previously published methods as density measures.¹⁵ We did not exclude periodic limb movement (PLM)-like muscle activity from RSWA scoring.^{15,31} Density measures including phasic, tonic, and any percent muscle activities were also calculated in the SM and in unilateral individual limb (FDS and AT) muscles, and bilateral limb phasic and any percent muscle activities were also calculated by counting each positive mini-epoch across both limbs (ie, by addition of all unique positive mini-epochs in each limb, whereas when both limbs contained a temporally simultaneous positive mini-epoch, only 1 positive mini-epoch was scored to avoid inflation of positive mini-epochs). In addition to these density metrics, the duration of each phasic burst during REM sleep was measured for each muscle and the average calculated for bilateral limb burst durations. Any 3-second mini-epoch containing a breathing-related, snoring-related, or spontaneous electroencephalography-defined arousal was excluded as artifact time from scoring and analysis.

Phasic muscle activity was defined as having an amplitude greater than 4 times the lowest background muscle activity voltage during REM sleep, with a duration between 0.1 and 14.9 seconds. Return of muscle activity to background for at least 200 ms delimited the end of each phasic burst. The percent phasic muscle activity as a measure of density was calculated based on the number of 3-second mini-epochs containing phasic muscle activity divided by the total number of REM sleep 3-second mini-epochs. Similarly, any percent muscle activity was calculated as the number of 3-second mini-epochs containing phasic muscle activity plus the number of 3-second mini-epochs containing tonic muscle activity divided by the total number of REM sleep 3-second mini-epochs (any 3-second mini-epoch containing both phasic and tonic muscle activity was only counted once).

All phasic density metrics were also calculated following AASM rules in 30-second epochs (for which a 30-second epoch is considered positive only when 5 or more 3-second mini-epochs contain phasic muscle activity).²⁷ We also used 30-second epochs to score tonic muscle activity in the SM, FDS, and AT muscles. An epoch was considered positive for tonic activity if $> 50\%$ of the epoch (ie, > 15 seconds in duration) had muscle activity continuously greater than double the

background EMG, or $> 10 \mu$ V. “Phasic-on-tonic” muscle activity was scored positively in addition to underlying tonic activity, as described before.³¹ All 4 scorers who visually quantified RSWA in this study demonstrated high interrater reliability, with a mean k-coefficient of 0.89 to our laboratory’s reference scoring, as previously described.¹⁵

The automated Ferri RAI for the SM muscle was calculated using HypnoLab sleep scoring software.¹⁴ Contrary to visually scored RSWA, when interpreting the automated RAI, lower values represent greater amounts of RSWA and higher values more normal levels of REM atonia (ie, an RAI of 0 would represent complete loss of REM sleep atonia and the highest amounts of RSWA possible, whereas an RAI of 1 represents completely preserved REM sleep atonia). Prior to RAI analysis, 30-second epochs containing a breathing-related artifact, snoring, or arousal were excluded, and the SM signal was notch-filtered at 60 Hz and rectified. We also calculated RAI in FDS and AT individual limb channels using the same methods for exclusion of physiologically increased REM sleep muscle activity, with particular caution to remove epochs with electrocardiogram or technical artifacts. When processed in this manner, the RAI essentially represents an integral function applied to the EMG signal, incorporating components of both density and duration.

In addition, we examined RSWA diagnostic thresholds and compared RSWA metrics between our 2017 cohort³¹ and this current cohort.

Statistical analysis

Clinical, demographic, and PSG data are presented as means and standard deviations. Categorical variables were analyzed with Fisher’s exact test or chi-square when applicable, while continuous variables were analyzed using nonparametric Mann-Whitney *U* test using R statistical software (R Foundation for Statistical Computing, Vienna, Austria). RSWA density scored in 3-second and AASM 30-second epochs were compared using Wilcoxon signed-rank tests.

A priori-defined co-primary outcome measures were the SM any and the combined SM+FDS any EMG activity, given that any muscle activity is the most parsimonious combined RSWA measure encompassing all types of muscle activity (phasic and/or tonic), and better specificity for SM and SM+FDS metrics for distinguishing RBD from controls in comparison to AT activity.^{15,31,34} All other RSWA measure comparisons were considered as secondary outcomes, including phasic, tonic, and any densities in individual or combined muscles; phasic muscle activity average burst durations; and RAI in chin and 4 limbs. A post hoc Bonferroni correction was applied for multiple tests for the SM any and SM + bilateral FDS any for comparisons between iRBD cases and controls, setting the experiment-wise alpha level at $P < .025$ for these a priori co-primary outcomes. For all secondary principal polysomnography and RSWA metric outcome comparisons, the alpha level was set at $P < .05$, so caution is advised in interpretation of these results given the possibility of type 1 error. Pearson correlation explored associations between RSWA metrics in the full cohort, and Spearman’s rank correlation was used to explore correlation between RSWA metrics and RBD symptom duration in patients with iRBD.

Receiver operating characteristic (ROC) curves were calculated for (1) phasic, tonic, and any percent muscle activities for SM, FDS, AT, and combined muscles; (2) phasic muscle burst durations; and (3) RAI. The area under the curve (AUC) was computed for each analysis, and 3 cutoff diagnostic threshold values were chosen yielding the following: (1) the highest combined sensitivity and specificity (optimized AUC cutoff), (2) maximized specificity, and (3) maximized sensitivity in distinguishing patients with iRBD from controls. To determine accurate cutoffs for clinical use, we calculated posttest likelihood ratios for several pretest likelihoods using RSWA metrics ROC curves.

Principal components analysis (PCA) defined determinants of RSWA metric variance and optimal muscles and/or muscle combinations distinguishing patients with iRBD from controls.

RESULTS

Demographic and polysomnographic data

Of the 20 patients with iRBD, 18 (90%) were men, with a mean age of 66.5 ± 7.7 years. Age, sex, and body mass index did not differ between patients and controls (Table 1). In the iRBD group, the mean duration of dream-enactment behavior symptom duration was 4.5 ± 3.6 (range 1–15) years. Principal sleep architecture metrics for the diagnostic study phases of PSGs

and SDB metrics for both diagnostic study phases and PSG condition for RSWA scoring were similar between groups, except for N3 sleep stage percentage, which was reduced in the iRBD group in comparison to controls, and a trend toward an overnight lower AHI in the iRBD group compared with controls.

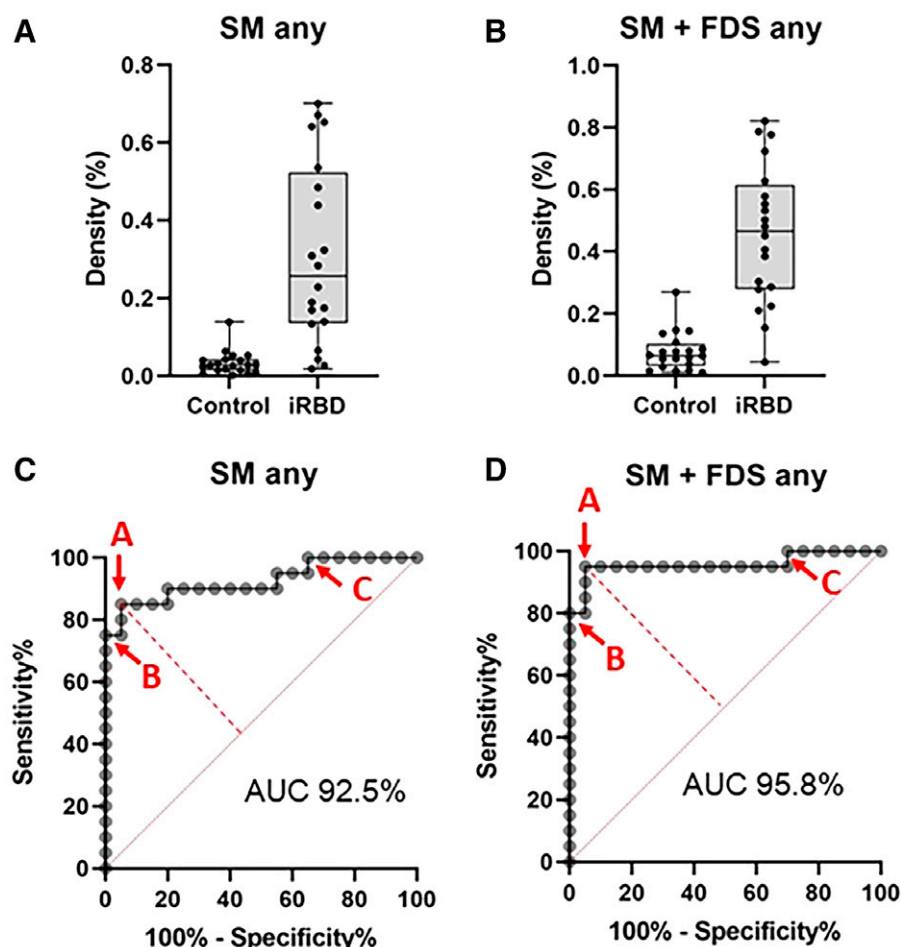
RSWA metrics and cutoffs

A mean total of 1.63% mini-epochs were excluded from analysis, with a similar amount of artifact exclusion between groups. Each of the co-primary outcomes for RSWA (SM any activity and combined SM+FDS any activity; ie, SINBAR montage) was greater in the iRBD group in comparison to controls ($P < .001$; Figure 1). As secondary outcomes, all SM (phasic and tonic), individual and bilateral FDS and AT (phasic, any), and combined muscle activity (ie, SM+FDS phasic, SM+AT, SM+FDS+AT) metrics were higher in the iRBD group (Table 2; all $P < .05$; Figure 2; complementary FDS and AT any metrics are shown in Table S1 in the supplemental material). Tonic activity in limbs was observed in 7 patients with iRBD and 2 controls (see Figure S1). Tonic muscle activity metrics in limbs did not differ between groups (Table S1). All phasic muscle activity burst duration variables, except for left and averaged AT phasic burst durations, were also longer in the iRBD group (Figure 2). The automated Ferri RAI performed in chin and 4 limbs also showed lower SM and each individual FDS values in patients with iRBD, while AT RAI did not differ between patients with iRBD and

Table 1—Patient demographics and polysomnogram metrics.

	iRBD	Controls	P
Patient demographics			
Age	66.5 ± 7.7	66.1 ± 10.2	.79
Sex (M/F)	18/2	16/4	.66
Body mass index, kg/m ²	27.8 ± 3.1	29.7 ± 6.7	.23
Principal diagnostic polysomnogram metrics			
Total sleep time, min	333.5 ± 65.3	342.2 ± 80.6	.28
Sleep efficiency %	73.7 ± 13.8	72.5 ± 15.7	.56
Stage N1%	13.0 ± 8.1	13.4 ± 7.5	.68
Stage N2%	53.6 ± 11.4	47.5 ± 13.1	.15
Stage N3%	13.6 ± 10.2	20.3 ± 10.2	.03
Stage REM %	19.8 ± 7.4	19.7 ± 5.8	.53
Initial sleep latency, min	16.2 ± 18.4	27.4 ± 78.5	.42
Initial REM latency, min	94.2 ± 48.2	96.4 ± 47.4	.45
Arousal index (/h)	19.9 ± 8.7	22.5 ± 12.7	.88
PLM index (/h)	34.9 ± 41.9	28.7 ± 30.3	.86
PLM with arousal index (/h)	3.7 ± 4.5	5.0 ± 7.1	.88
Apnea-hypopnea index total (events/h)	4.4 ± 4.7	8.2 ± 8.2	.07
Respiratory metrics in condition scored*			
Apnea-hypopnea index (events/h)	3.1 ± 3.8	3.9 ± 4.6	.88
REM apnea-hypopnea index (/h)	1.3 ± 1.6	3.6 ± 4.5	.14

*The respiratory metrics corresponding to either the diagnostic or treatment section of the polysomnogram during which RSWA was selected and scored. iRBD = isolated rapid eye movement sleep behavior disorder, M/F = male/female, PLM = periodic limb movements, REM = rapid eye movement, RSWA = rapid eye movement sleep without atonia.

Figure 1—Boxplot representations and receiver operating characteristic curves of SM any and SM+FDS any.

Boxplot representations of SM any (A) and SM+FDS any (B) muscle activity densities measured in 3-second mini-epochs in controls and patients with iRBD. Corresponding receiver operating characteristic curves of SM any (C) and SM+FDS any (D) showing 3 cutoffs—A (in red): optimized cutoff (yielding highest combined sensitivity and specificity); B (in red): maximized specificity; C (in red): maximized sensitivity. AUC = area under the curve, FDS = flexor digitorum superficialis, iRBD = isolated rapid eye movement sleep behavior disorder, SM = submental muscle.

controls (Figure 2). We were not able to perform RAI in both FDS in 1 patient with iRBD and 1 control, and in both AT in 1 patient with iRBD and 1 control because of electrocardiogram artifacts in limb muscles. All RAI and visually scored any, tonic, and phasic metrics showed a strong negative correlation in SM, FDS, and AT (Table S2), except for tonic in the left AT muscle. The PLM index (PLMI) was not correlated with combined phasic density in the AT. Duration of symptoms was uncorrelated with any of the primary outcome measures in patients with iRBD (SM any: $P = .60$; SM+FDS any: $P = .87$).

ROC analyses determined optimized AUC diagnostic cutoffs in 3-second mini-epochs: SM any = 6.5% and combined SM + bilateral FDS any = 15.1%, yielding a specificity of 95% and sensitivities of 85% and 95%, respectively (Table 3 and Figure 1). To be used in various clinical settings, we also defined cutoffs with maximized specificity (SM any = 14.0% and combined SM+FDS any = 27.4%) and maximized sensitivity (SM any = 1.8% and combined SM+FDS any = 4.1%).

Those 3 cutoffs were also calculated for each type of activity in individual muscles, combined muscles, phasic burst activity durations, and RAI metrics that were different between patients with iRBD and controls (Table 2, Figure 2). The greatest AUC values were found for SM+FDS any and phasic, combined, and individual FDS phasic or any; left and averaged FDS RAI, SM any, phasic burst durations and RAI; and combined SM+FDS+AT any. However, AUC comparisons did not show significant differences between SM any and SM+FDS any ($P = .57$), and between SM RAI and averaged FDS RAI ($P = .83$).⁴⁴ To calculate likelihood ratios of RSWA cutoffs, we used pretest likelihoods of 25%, 50%, and 75% in patients with a history of dream-enactment behavior. The optimized AUC cutoffs for SM any, SM+FDS any, SM+FDS+AT any, combined FDS any, and averaged FDS RAI displayed higher posttest likelihoods for all 3 pretest likelihood levels of iRBD than maximized specificity cutoffs (Table S3). For an incidental finding of RSWA without dream-enactment behavior, we used pretest likelihoods of 2%, 5%, and 10% according to prevalence studies in

Table 2—RSWA analysis in patients with iRBD and controls in 3-second mini-epoch scoring.

3-Second Scoring	iRBD	Controls	P
% Mini-epochs excluded	1.68 ± 0.88	1.58 ± 1.13	.56
Mini-epochs analyzed	1,085.5 ± 541.8	907.2 ± 405.3	.31
Coprimary outcomes			
SM any %	31.2 ± 23.2	3.3 ± 3.0	<.001
SM+FDS any %	45.6 ± 22.3	7.7 ± 6.2	<.001
Secondary outcomes			
SM phasic %	19.7 ± 13.5	3.3 ± 3.0	<.001
SM tonic %	11.7 ± 16.9	0.07 ± 0.3	<.001
Combined FDS phasic %	24.8 ± 11.9	4.5 ± 4.7	<.001
Left FDS phasic %	14.8 ± 8.0	2.5 ± 3.3	<.001
Right FDS phasic %	16.3 ± 8.9	3.0 ± 3.9	<.001
Combined FDS any %	24.9 ± 11.9	4.6 ± 4.8	<.001
Combined AT phasic %	24.2 ± 14.3	13.7 ± 18.9	.001
Left AT phasic %	13.5 ± 8.5	7.6 ± 8.8	.006
Right AT phasic %	16.2 ± 12.9	9.2 ± 17.0	.002
Combined AT any %	24.2 ± 14.3	13.7 ± 18.9	.001
SM+FDS phasic %	37.6 ± 16.3	7.6 ± 6.0	<.001
SM+AT phasic %	38.4 ± 15.0	16.8 ± 18.7	<.001
SM+AT any %	46.5 ± 20.9	16.8 ± 18.7	<.001
SM+FDS+AT phasic %	49.9 ± 16.4	19.7 ± 19.5	<.001
SM+FDS+AT any %	55.8 ± 20.1	19.8 ± 19.5	<.001
Burst durations			
SM duration, seconds	1.06 ± 0.44	0.53 ± 0.48	<.001
Left FDS duration, seconds	0.76 ± 0.25	0.48 ± 0.35	.001
Right FDS duration, seconds	0.79 ± 0.36	0.39 ± 0.19	<.001
Averaged FDS duration, seconds	0.77 ± 0.29	0.47 ± 0.33	<.001
Left AT duration, seconds	0.50 ± 0.21	0.45 ± 0.39	.09
Right AT duration, seconds	0.57 ± 0.26	0.38 ± 0.19	.01
Averaged AT duration, seconds	0.54 ± 0.21	0.45 ± 0.35	.05
Automated RSWA (RAI) measures			
SM RAI	0.714 ± 0.228	0.956 ± 0.028	<.001
Left FDS RAI	0.829 ± 0.132	0.964 ± 0.024	<.001
Right FDS RAI	0.822 ± 0.195	0.962 ± 0.028	<.001
Averaged bilateral FDS RAI	0.828 ± 0.156	0.961 ± 0.027	<.001
Left AT RAI	0.864 ± 0.104	0.924 ± 0.057	.08
Right AT RAI	0.831 ± 0.151	0.899 ± 0.116	.06
Averaged bilateral AT RAI	0.853 ± 0.119	0.913 ± 0.079	.07

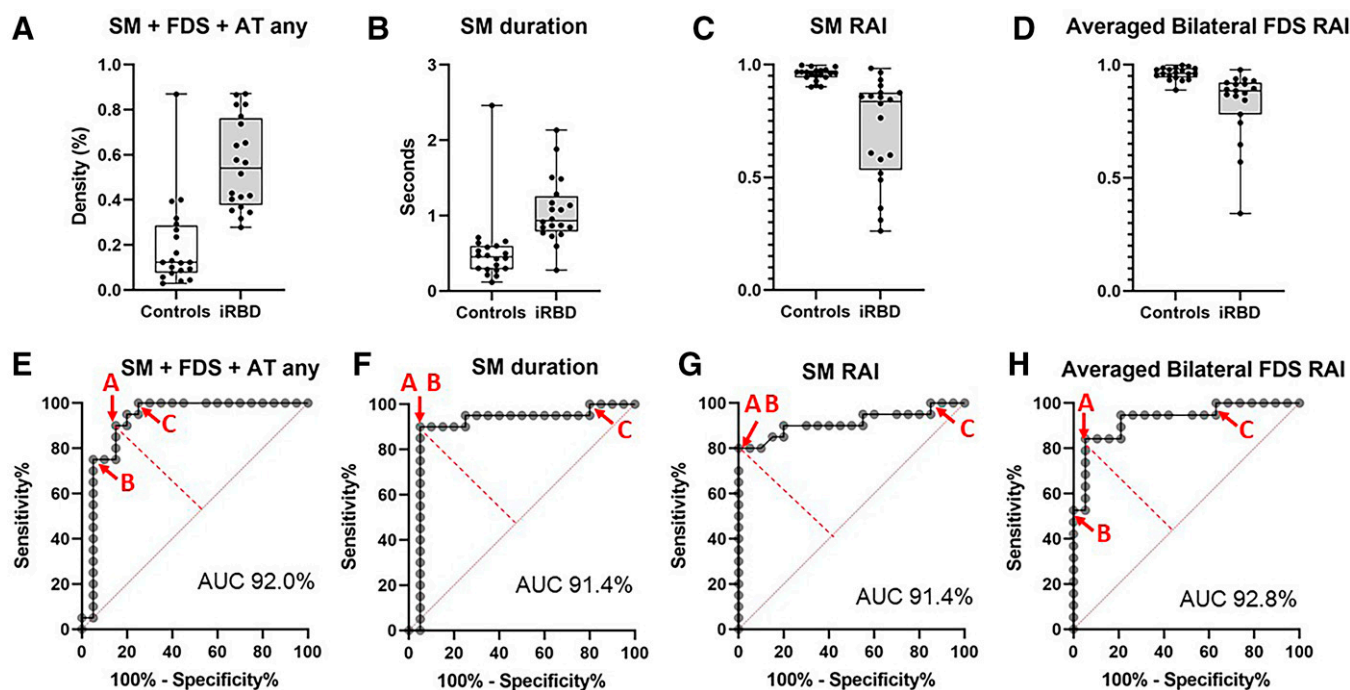
AT = anterior tibialis, FDS = flexor digitorum superficialis, iRBD = isolated rapid eye movement sleep behavior disorder, RAI = REM atonia index, RSWA = rapid eye movement sleep without atonia, SM = submental muscle.

community, older, and sleep laboratory–biased populations.^{45,46} The maximized specificity AUC cutoffs resulted in more conservative predictions in SM any, SM+FDS any, combined FDS any, and averaged FDS RAI, with a 4–36% chance that an elevated measure would represent a true marker of iRBD (**Table S3**).

We performed PCA for all visually scored RSWA metrics in order to define determinants of RSWA metric variance, and those muscles and/or muscle combinations that best distinguished

patients with iRBD from controls (**Figure 3** and **Table S4**). Principal component (PC) 1 clustered patients with iRBD according to combined FDS any and phasic, SM+AT any and phasic, and SM+FDS any and phasic RSWA, with a much smaller proportion of variance attributable to PC2. Controls were best represented by PC1 and PC2, which positively correlated with AT any and phasic density and negatively with FDS duration, SM any and phasic, and AT duration.

Figure 2—Boxplot representations and corresponding receiver operating characteristic curves of secondary outcome SM+FDS+AT any, SM duration, SM RAI and averaged bilateral FDS RAI.



Boxplot representations of combined SM+FDS+AT any muscle activity densities measured in 3-second mini-epochs (A), SM duration of muscle burst activities (B), automated SM RAI (C) and averaged bilateral FDS RAI (D) in controls and patients with iRBD. (E–H). Corresponding receiver operating characteristic curves showing 3 cutoffs—A (in red): optimized cutoff (yielding highest combined sensitivity and specificity); B (in red): maximized specificity; C (in red): maximized sensitivity. AT = anterior tibialis, AUC = area under the curve, FDS = flexor digitorum superficialis, iRBD = isolated rapid eye movement sleep behavior disorder, RAI = REM atonia index, SM = submental muscle.

We compared these current cohort RSWA data with that of our 2017 cohort³¹ (Table S5). Symptom duration did not differ significantly between cohorts (5.7 ± 3.1 [range 1–10] years in the 2017 cohort, $P = .20$). SM any, SM tonic, combined AT phasic, and SM RAI were similar between 2017 and 2022 cohorts, whereas SM phasic and SM+AT phasic were lower in the 2022 cohort than in the 2017 cohort ($P = .02$ and $P = .001$, respectively; Table S5). However, when adjusting diagnostic cutoff determination with similar specificity levels on ROC analysis, SM phasic and SM+AT phasic exhibited similar cutoff values (eg, SM phasic cutoff of 15.8% in 2017 vs 15.2% in 2022, with 100% specificity).

In order to provide metrics and thresholds adapted to current AASM scoring criteria for RSWA, we performed group comparisons with AASM 30-second epoch scoring, which showed higher RSWA metrics values in all individual or combined muscles in patients with iRBD in comparison to controls (Table 4). In controls, comparison between AASM metrics and 3-second metrics yielded differences for every individual muscle or combined muscles (all $P < .01$). In patients with iRBD, all metrics differed between AASM and 3-second scoring (all $P < .02$), except for SM+AT phasic density ($P = .08$) and SM+AT+FDS phasic density ($P = .93$). We also performed ROC curves in AASM 30-second metrics and proposed cutoffs with optimized

sensitivity and specificity, maximized specificity, and maximized sensitivity (Table 5).

DISCUSSION

We confirmed that visually determined phasic, tonic, and any muscle activity percentages reflecting densities were higher in patients with iRBD than in controls, and further validated that phasic activity burst durations are an accurate diagnostic measure for iRBD. We also determined 3 useful diagnostic RSWA cutoffs for chin and 4 limbs metrics with optimized specificity and sensitivity, maximized specificity, and maximized sensitivity applicable in different clinical settings, depending on diagnostic confirmation or detection needs having varying levels of clinical suspicion and confidence. In addition, we confirmed the value of the automatic Ferri RAI RSWA measure in FDS, which yielded a strong correlation with visually scored 3-second RSWA metrics and a large AUC in ROC analyses. PCA confirmed the importance of FDS and SM activity (density and duration) in classifying patients with iRBD, while AT density partially drove controls clustering, suggesting that it is less specific and discriminative for accurate iRBD diagnosis.

Table 3—Cutoff values for RBD diagnosis in 3-second mini-epoch scoring.

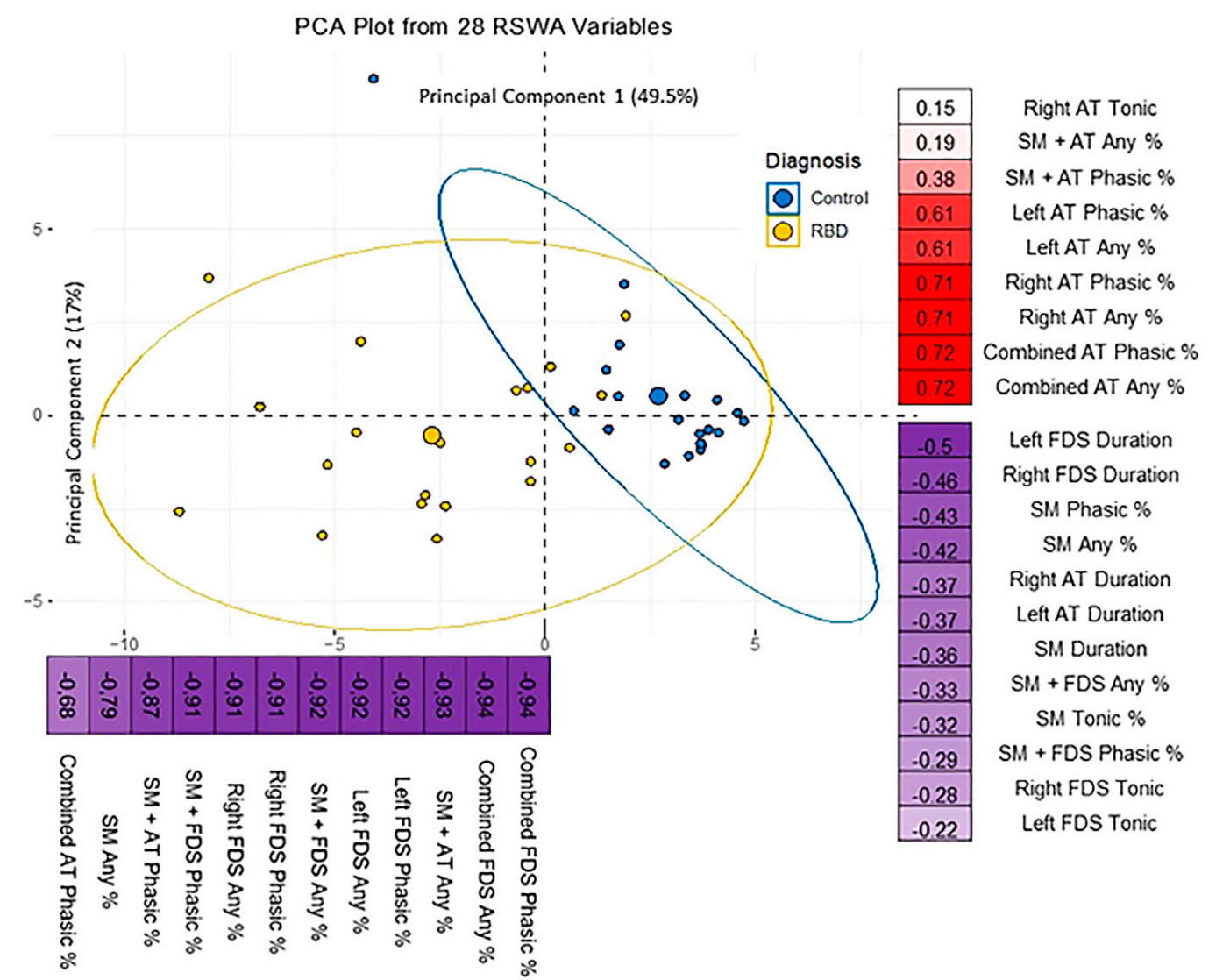
	Optimized AUC	Specificity	Sensitivity	AUC	Maximized Specificity	Maximized Sensitivity
Co-primary metrics (density)						
SM any %	6.5	95.0%	85.0%	92.5%	14.0	1.8
SM+FDS any %	15.1	95.0%	95.0%	95.8%	27.4	4.1
Muscle combinations (density)						
SM+FDS phasic %	15.0	95.0%	95.0%	95.5%	27.2	4.1
SM+AT phasic %	16.5	70.0%	100.0%	87.5%	39.3	16.5
SM+AT any %	21.0	75.0%	95.0%	88.5%	40.4	17.6
SM+FDS+AT phasic %	26.4	75.0%	100.0%	91.3%	40.2	26.4
SM+FDS+AT any	33.1	85.0%	90.0%	92.0%	86.9	27.2
Combined FDS phasic %	10.7	90.0%	95.0%	96.0%	15.3	2.6
Combined FDS any %	11.6	90.0%	90.0%	95.8%	15.3	2.6
Combined AT phasic %	7.7	55.0%	100.0%	78.8%	39.6	7.7
Combined AT any %	7.7	55.0%	100.0%	78.8%	39.6	7.7
Individual muscles (density)						
SM phasic %	6.5	95.0%	85.0%	91.2%	15.1	1.8
SM tonic %	0.5	95.0%	70.0%	83.2%	1.8	0.5
Left FDS phasic %	7.3	95.0%	90.0%	95.5%	13.9	1.9
Left FDS any %	7.3	95.0%	90.0%	95.5%	13.9	1.9
Right FDS phasic %	5.4	85.0%	95.0%	93.5%	14.1	0.8
Right FDS any %	5.8	85.0%	90.0%	93.2%	14.1	0.8
Left AT phasic %	4.8	50.0%	95.0%	75.2%	16.8	2.1
Left AT any %	4.8	50.0%	95.0%	75.8%	16.8	2.1
Right AT phasic %	4.7	60.0%	90.0%	78.2%	31.1	2.8
Right AT any %	4.7	60.0%	90.0%	78.2%	31.5	2.8
Phasic muscle activity burst durations						
SM duration, s	0.72	95.0%	90.0%	90.2%	0.72	0.27
Averaged FDS duration, s	0.46	70%	85%	81.5%	0.98	0.29
Left FDS duration, s	0.45	70.0%	95.0%	82.2%	1.04	0.30
Right FDS duration, s	0.50	90.0%	70.0%	85.2%	0.91	0.23
Averaged AT duration, s	0.36	60%	80%	68%	0.96	0.24
Left AT duration, s	0.32	60.0%	80.0%	66.0%	0.96	0.17
Right AT duration, s	0.41	70.0%	80.0%	78.2%	0.83	0.25
Automated measures						
SM RAI	0.888	100.0%	80.0%	91.4%	0.888	0.988
Averaged bilateral FDS RAI	0.930	94.7%	84.2%	92.8%	0.887	0.978
Left FDS RAI	0.930	94.1%	94.4%	96.7%	0.914	0.966
Right FDS RAI	0.940	84.2%	84.2%	88.6%	0.879	0.997

Optimized area under the curve (AUC) correspond to cutoffs yielding the highest combined sensitivity and specificity. Cutoffs with maximized specificity are provided before last column, and those with maximized sensitivity in the last column. [Figure 1](#), [Figure 2](#), and [Figure 3](#) illustrate the 3 cutoffs determination method. AT = anterior tibialis, FDS = flexor digitorum superficialis, RAI = REM atonia index, RBD = rapid eye movement sleep behavior disorder, SM = submental muscle.

Our results confirm previous studies from our team and others and point toward certain RSWA metrics that appear particularly robust and reproducible across different patient cohorts, time points, and visual and automated scoring methodology used. We determined that a density of 14% in SM any

and 15.1% in SM phasic exhibited maximized specificity, which is close to 19.7% and 15.8% in our 2017 cohort³¹ (optimized cutoff, but also maximized specificity) and 18.2% and 16.3% in the Frauscher et al study³⁴ from 2012. Despite some differences in RSWA scoring methods, we also obtained similar

Figure 3—Graphical representation of PCA performed on 28 visually scored RSWA parameters (first 12 for principal component [PC] 1 and 2 are listed in the graphic).



Patients with iRBD are gathered according to PC1 mainly (parameters influencing variance are listed in decreasing order with highest negative Pearson coefficient close to the central vertical line, decreasing toward the left), with a much smaller proportion of variance attributable to PC2. Controls are best classified by PC1 and PC2, correlated negatively with FDS burst durations (parameters listed in decreasing order from central horizontal line to the bottom) and positively with AT densities (parameters in decreasing order from central horizontal line to the top). Parameters followed by % refer to densities. Full parameters and Pearson coefficients are listed in **Table S4**. AT = anterior tibialis, FDS = flexor digitorum superficialis, iRBD = isolated rapid eye movement sleep behavior disorder, PCA = principal components analysis, RBD = rapid eye movement sleep behavior disorder, RSWA = rapid eye movement sleep without atonia, SM = submental muscle.

results for our maximized specificity cutoffs for density in comparison to the SINBAR group: in SM+FDS phasic (27.2% in the current cohort vs 25.3% in 2012), SM+AT phasic (39.3% in 2022 vs 44.2% in 2012), combined AT phasic (39.6% in 2022 vs 39.2% in 2012), and combined FDS phasic (15.3% in 2022 vs 16.8% in 2012). We also provided group comparisons and RSWA cutoffs following current AASM-recommended scoring methods of tonic and phasic activity in individual and combined muscles,²⁷ which are potentially useful when using the AASM rules as 30-second and 3-second scoring methods yielded significantly different measures in patients with iRBD and

controls. While 3-second and AASM 30-second scoring density thresholds showed similar accuracy overall, artifact discrimination is not considered in AASM rules, which may be particularly problematic in the setting of severe REM-related obstructive sleep apnea or frequent spontaneous arousals during REM sleep. The AASM method using 30-second epochs also had lower sensitivity for limb RSWA measures than 3-second mini-epoch scoring, leading to under-recognition of abnormal phasic activity.^{15,31} Future studies in larger cohorts should continue to comparatively analyze the diagnostic accuracy parameters between 3-second and 30-second epoch scoring

Table 4—RSWA analysis in patients with iRBD and controls following AASM scoring methods.

AASM 30-Second Scoring	iRBD	Controls	P
Muscle combinations			
SM+FDS phasic %	32.9 ± 21.8	2.3 ± 4.1	<.001
SM+AT phasic %	33.8 ± 24.8	12.2 ± 23.1	<.001
SM+FDS+AT phasic %	49.8 ± 26.0	14.3 ± 23.7	<.001
Combined FDS phasic %	17.8 ± 14.4	0.6 ± 1.3	<.001
Combined AT phasic %	17.3 ± 21.6	10.3 ± 22.9	.02
Individual muscles			
Phasic SM %	12.0 ± 14.7	0.5 ± 1.0	<.001
Phasic left FDS %	7.1 ± 6.9	0.2 ± 0.7	<.001
Phasic right FDS %	7.3 ± 7.7	0.4 ± 1.2	<.001
Phasic left AT %	5.3 ± 7.5	2.8 ± 7.6	.03
Phasic right AT %	10.1 ± 17.0	6.9 ± 21.2	.01

AASM = American Academy of Sleep Medicine, AT = anterior tibialis, FDS = flexor digitorum superficialis, iRBD = isolated rapid eye movement sleep behavior disorder, RSWA = rapid eye movement sleep without atonia, SM = submental muscle.

approaches to determine the relative utility of these different methods in various clinical applications.

As previously suggested,^{13,34,35} our PCA highlighted the importance of FDS activity in classifying patients with iRBD vs controls, followed by chin activity and combined measures. In comparison, AT activity alone was poorly discriminant in classifying patients with iRBD vs controls. Of note, the lower specificity of AT density was not fully explained by PLMI in our population, as these measures were not correlated. Including or excluding PLM-like movements in RSWA scoring does not seem to modify the AT density specificity, as AT was also one of the lowest discriminant muscles in a SINBAR study excluding PLM-like movements from RSWA scoring.³⁴ We also found that

RSWA diagnostic thresholds for density were similar between our own current cohort vs 2017 iRBD cohort, although a few RSWA measures were lower than in our 2017 cohort. Future studies should continue to analyze whether there are any emerging temporal trends in RSWA diagnostic thresholds, as enhanced awareness among sleep clinicians of the potential value of visually observed RSWA would indeed raise such a possibility.

The Ferri RAI was initially developed for iRBD diagnosis in the chin muscle,¹⁴ and is thought to reflect both phasic and tonic muscle activity. However, we showed strong correlations between the automated RAI and manually scored any RSWA metrics in SM, as well as any and phasic metrics in limbs, possibly higher than with tonic activity. We also confirmed that FDS

Table 5—Cutoff values for RBD diagnosis following AASM scoring methods.

AASM 30-Second Scoring	Optimized AUC	Specificity	Sensitivity	AUC	Maximized Specificity	Maximized Sensitivity
Muscle combinations						
SM+FDS phasic	6.2%	95.0%	95.0%	95.5%	21.9%	0.3%
SM+AT phasic	18.7%	90.0%	70.0%	83.3%	46.6%	3.7%
SM+FDS+AT phasic	23.8%	85.0%	85.0%	88.4%	43.3%	10.2%
Combined FDS phasic	5.3%	100%	90.0%	96.4%	5.3%	2.0%
Combined AT phasic	13.9%	85%	50%	71.1%	52.2%	1.3%
Individual muscles						
SM phasic	2.3%	95.0%	70.0%	88.6%	4.2%	0.3%
Left FDS phasic	1.8%	95.0%	85.0%	93.5%	2.7%	0.9%
Right FDS phasic	2.0%	95.0%	80.0%	89.8%	6.0%	0.7%
Left AT phasic	1.3%	70.0%	65.0%	69.5%	8.2%	0.3%
Right AT phasic	2.2%	75.0%	70.0%	72.4%	23.0%	0.4%

AASM = American Academy of Sleep Medicine, AT = anterior tibialis, AUC = area under the curve, FDS = flexor digitorum superficialis, RBD = rapid eye movement sleep behavior disorder, SM = submental muscle.

RAI is a useful measure in iRBD diagnosis, with a 0.930 cutoff demonstrating 94.7% specificity and 84.2% sensitivity.³⁵

Chin and 4 limbs duration of phasic muscle activity bursts also had utility in the accurate classification of patients with iRBD vs controls in our PCA, and SM duration ROC curve showed a particularly high performance, with 0.72-second average duration of SM phasic bursts yielding 95% specificity and 90.2% sensitivity (AUC 90.2%). This threshold of approximately 0.7–1.0 seconds for SM duration appears robust across several studies involving different prodromal and manifest synucleinopathy patients as well as normative populations from our group (previously, 0.65 seconds and 0.66 seconds in iRBD,^{15,31} 0.78 seconds in iRBD nonconverters, and 0.94 seconds in iRBD phenoconverters³²; 0.64 seconds and 0.65 seconds for distinguishing probable synucleinopathy from tauopathy cognitive and motor phenotype patients^{5,33}; and 1.1 seconds representing the 95th percentile in a normative population of 118 adults throughout the lifespan).³⁰ In the visual interpretation of RSWA in clinical practice, paying particular attention to the average RSWA phasic burst duration may serve as a shorthand screening method to aid in iRBD diagnosis, with frequent long SM burst durations (ie, of ~0.7–1.0 seconds duration) being highly suggestive of abnormally excessive RSWA levels and the need for more detailed quantitative analysis for accurate diagnosis.

Similarly, since clinical polysomnographers typically analyze all available EMG data when interpreting RSWA qualitatively in the sleep laboratory, we provided diagnostic thresholds for combined SM+FDS+AT activity to guide the overall visual estimation of the RSWA scorer when considering all EMG activity across channels, finding that 26.4% for combined phasic muscle activity across channels or 33.1% for the parsimonious (“all in”) RSWA measure of any muscle activity across channels were accurate for iRBD diagnosis, which are convenient, memorable percentages for 3-second mini-epoch scoring applicable in practice settings.

Which cutoffs should we consider for use in clinical practice?

Isolated RSWA identified incidentally on PSG is a recent concept that may underlie prodromal RBD, in which patients exhibit excessive RSWA without behavioral events or, alternatively, isolated RSWA may identify possible prodromal alpha-synucleinopathy patients without dream enactment that may follow an alternative trajectory to eventually manifest Parkinson disease, dementia with Lewy bodies, or multisystem atrophy. Isolated RSWA frequency has been reported to be 14–32% in sleep laboratory and community populations,^{30,47} and follow-up demonstrates a likely association with underlying neurodegeneration, with 7–14% of patients with isolated RSWA in 1 longitudinally followed cohort converting to iRBD, accompanied by 71% positivity for neurodegenerative biomarker presence.⁷ For research purposes in naturalistic studies, we proposed the use of normative 95th percentile RSWA thresholds to investigate the role of isolated RSWA in neurodegeneration.³⁰ However, due to ethical implications, one should avoid overestimating excessive RSWA in an asymptomatic patient; thus, we suggest applying maximized specificity cutoffs for iRBD in patients with putative isolated RSWA in the general population in practice.

The clinical setting in suspected iRBD is different, as patients usually display overt dream enactment with potentially injurious behaviors. When considering this setting of an overt clinical presentation, using maximized specificity cutoffs or optimized cutoffs remains debatable.^{31,34} Ethical pitfalls in over-diagnosing iRBD should be considered, as well as underestimating iRBD diagnosis from the perspective of candidacy for inclusion in future neuroprotective trials.^{48,49} Due to better diagnostic performance (accurate likelihood ratios) in patients with suspected iRBD, we would recommend using optimized AUC cutoffs in the setting of clinically suspected iRBD. RSWA quantification has also been shown to enable distinction of probable synucleinopathies from tauopathies in patients presenting with parkinsonism and in elderly adults with cognitive impairment.^{5,33} Moreover, due to its high prevalence in synucleinopathies, such as multisystem atrophy and dementia with Lewy bodies, the absence of RBD or RSWA appears as an atypical feature that may argue against a synucleinopathy diagnosis. Maximized sensitivity or optimized cutoffs could then be useful in these neurodegenerative populations to assess paucisymptomatic RBD or isolated RSWA with greater sensitivity.

Our study has several limitations. RSWA was an inclusion criterion for the iRBD population and an outcome measure, which can be seen as a circular issue; however, inclusion criteria were a clinical history of dream-enactment behaviors and qualitatively determined RSWA presence as interpreted by a board-certified sleep neurologist, which is in accordance with ICSD-3 criteria.¹ Our patients with iRBD and controls are from a single tertiary center and may not reflect the population seen in sleep centers with different referral practice and population profiles. Controls were not healthy individuals, but patients referred for SDB whose PSG demonstrated a low AHI in the condition in which RSWA was scored. This selection bias is a limitation, yet is naturalistic since SDB controls are likely to be similar and representative of real-world populations encountered in sleep centers that require distinction from iRBD. Eight (40%) patients with iRBD and 13 (65%) controls underwent split-night sleep studies in which we selected the second half for RSWA scoring; however, split-night sleep studies were equally distributed between groups; moreover, tonic and phasic activity appears to be stable over nighttime.⁵⁰ Finally, we had a few missing data in automated (RAI) measures in limbs because of electrocardiogram artifacts. As RAI performed in FDS appears to be a discriminating parameter, this automated measurement emphasizes the need for flawless technical quality to ensure its feasibility and applicability.

Future developments in RSWA analysis tend toward automated and thus more time-efficient, standardized, and less labor-intensive methods. However, currently available automated approaches to RSWA scoring still require manual removal of artifacts and muscle activities related to respiratory events and arousals,⁵¹ and rely mainly on chin muscle activity, rarely considering limb muscle activity, and requiring the use of proprietary commercial softwares.^{14,19,25,26,51–53} Cesari et al⁵¹ performed a systematic comparison of 5 algorithms in different cohorts and did not report an optimal method, except that the RAI exhibited the highest sensitivity. Those 5 methods were not influenced by movements related to apneas and arousals, making them promising for eventual applications toward clinical populations. Automated and open procedures could help extend systematic quantification of RSWA in clinical

practice to improve patients' evaluation and harmonize research procedures. To facilitate comprehensive, state-of-the art diagnosis, prognosis, and timely therapy for iRBD and other prodromal synucleinopathies, biomarker development and eventual neuroprotective trials will require cooperative multicenter initiatives with standardized and harmonized clinical and neurophysiological assessments, such as those in the North American Prodromal Synucleinopathy (NAPS) Consortium (<https://www.naps-rbd.org>, <https://clinicaltrials.gov/ct2/show/NCT03623672>).⁵⁴

In conclusion, we determined contemporary RSWA cutoffs in chin and 4 limb muscles applicable in current iRBD populations, considering uses in different clinical situations and the need for adjusted sensitivity and/or specificity accordingly. We confirmed the key contribution of SM or FDS activity in accurately classifying patients with iRBD vs controls and the value of a time-efficient automated measure such as the Ferri RAI performed in SM or FDS; we also found that, in most instances, SM any muscle activity evaluation might suffice given its relatively comparable diagnostic performance to SM+FDS or FDS alone. The possibility that earlier RBD diagnosis could lead to milder phenotypes and lower RSWA levels at presentation suggests the need for continued monitoring of RSWA thresholds to enable accurate iRBD diagnosis and application of eventual neuroprotective treatments to delay or prevent the eventual devastating neurodegenerative cognitive, motor, and autonomic consequences of overt synucleinopathies including Parkinson disease, dementia with Lewy bodies, and multiple system atrophy.

ABBREVIATIONS

AASM, American Academy of Sleep Medicine
 AT, anterior tibialis
 AUC, area under the curve
 EMG, electromyography
 FDS, flexor digitorum superficialis
 iRBD, isolated rapid eye movement sleep behavior disorder
 PC, principal component
 PCA, principal components analysis
 PLM, periodic limb movement
 PSG, polysomnographic/polysomnography
 RAI, REM atonia index
 REM, rapid eye movement
 ROC, receiver operating characteristic
 RSWA, rapid eye movement sleep without atonia
 SDB, sleep-disordered breathing
 SINBAR, Sleep Innsbruck Barcelona
 SM, submental muscle

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